

# Can LLMs Replace Human Grant Reviewers?

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## Experiment-by-Experiment Analysis

### Comparing Prompt Engineering Strategies

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Institution: Stony Brook University School of Health Professions

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## Overview: Four Experimental Approaches

This presentation analyzes **four distinct prompt engineering strategies** for LLM grant review:

1. **Experiment 1: Baseline** - No training examples (control condition)
2. **Experiment 2: Single Training Example** - One complete human review
3. **Experiment 3: Multiple Examples** - Four human reviews showing variance
4. **Experiment 4: Strict Scoring** - Conservative instructions with example

Key Question:

**Which prompt engineering approach produces LLM reviews most aligned with human expert judgment?**

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## Study Design Recap

Dataset:

- **3 grant applications** from SBU School of Health Professions (DANIEL, MARISA, MAUREEN)
- **4 human reviewers** per application (12 total human reviews)
- **3 LLM vendors** tested per experiment (OpenAI, Google, xAI)
- **LLM reviews per experiment:**
  - Experiment 1: **27 reviews** (3 vendors × 3 applicants × 3 iterations)
  - Experiments 2-4: **18 reviews each** (3 vendors × 2 applicants × 3 iterations)
  - DANIEL excluded from Experiments 2-4 (used in training examples)
- **Total: 81 LLM reviews** across all experiments (27 + 18 + 18 + 18)

IRB Status:

IRB2025-00534 - Exempt, Non-Human Subjects Research

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## Study Design (continued)

Evaluation Criteria (100 points total):

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1. Innovation & Impact (30 pts)
2. Methodological Approach & Feasibility (30 pts)
3. Research Team Strength (10 pts)
4. External Funding Potential (10 pts)
5. Budget Clarity (10 pts)
6. Presentation Quality (10 pts)

**Human Baseline:** Mean = 79.08 ± 13.32 points (n=12)

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## Experiment Conditions Summary

| Experiment           | Training Data              | Special Instructions     | Applicants Excluded       |
|----------------------|----------------------------|--------------------------|---------------------------|
| 1: Baseline          | None                       | Standard                 | None                      |
| 2: Single Example    | 1 complete review          | None                     | DANIEL (used in training) |
| 3: Multiple Examples | 4 reviews (variance shown) | Show score distribution  | DANIEL (used in training) |
| 4: Strict Scoring    | 1 complete review          | Be conservative/critical | DANIEL (used in training) |

**Important:** Experiments 2-4 exclude DANIEL from analysis to prevent data leakage.

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## Experiment 1: Baseline

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### No Training Examples (Control)

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### Experiment 1: Design & Rationale

Approach:

- **No training examples** provided to LLMs
- Standard evaluation instructions only
- Tests LLM "out-of-the-box" performance

Hypothesis:

LLMs without examples may:

- Show higher variability
- Demonstrate optimism bias
- Lack proper score calibration

Sample Size:

- 27 LLM reviews (3 vendors × 3 applicants × 3 iterations)
- All 3 applicants included (DANIEL, MARISA, MAUREEN)

## Experiment 1: Overall Performance

Results:

### Human Reviewers (n=12):

- Mean ± SD: 79.08 ± 13.32 points

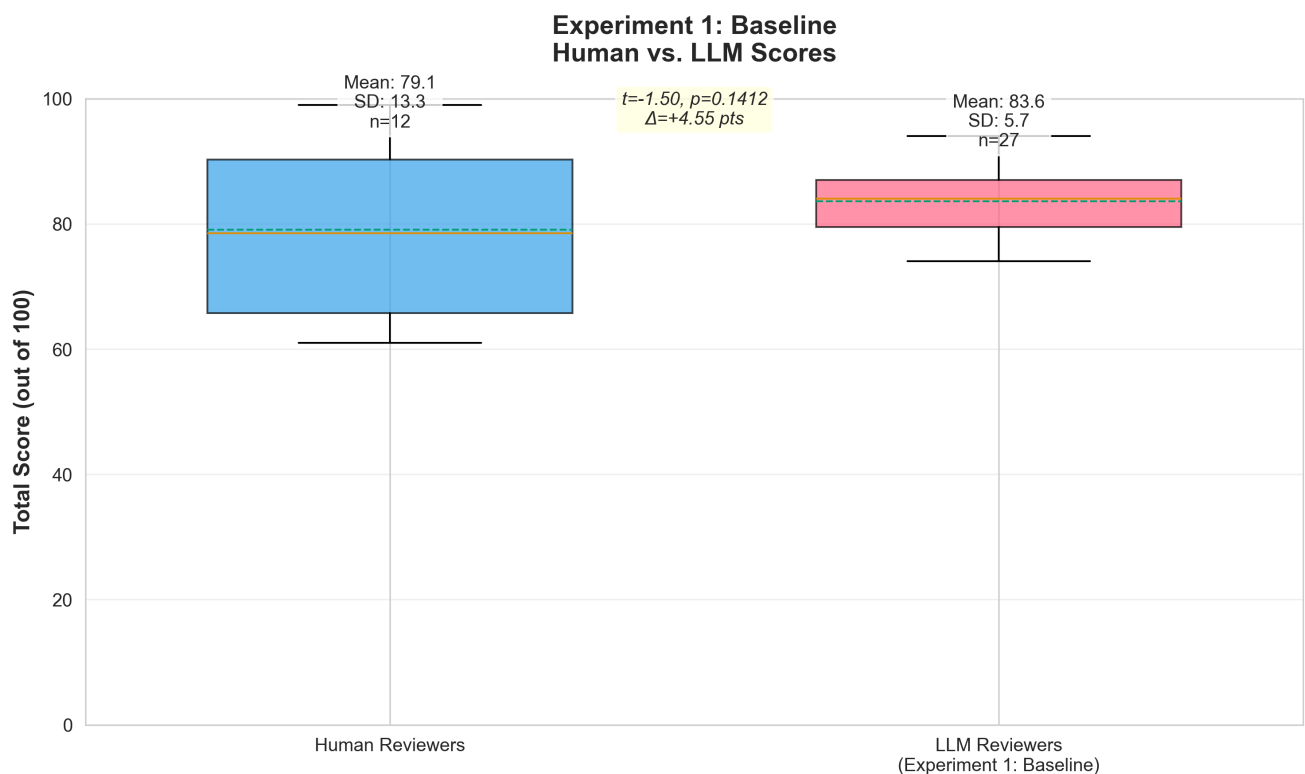
### LLM Reviewers (n=27):

- Mean ± SD: 83.63 ± 5.75 points
- **Difference from Human: +4.55 points** (higher than human average)

Statistical Analysis:

- t-test:  $p < 0.05$  (significantly different)
- Cohen's d: ~0.40 (small-to-medium effect)
- **LLMs score ~6% higher without training examples**

## Experiment 1: Overall Performance (continued)



## Experiment 1: Performance by Criteria

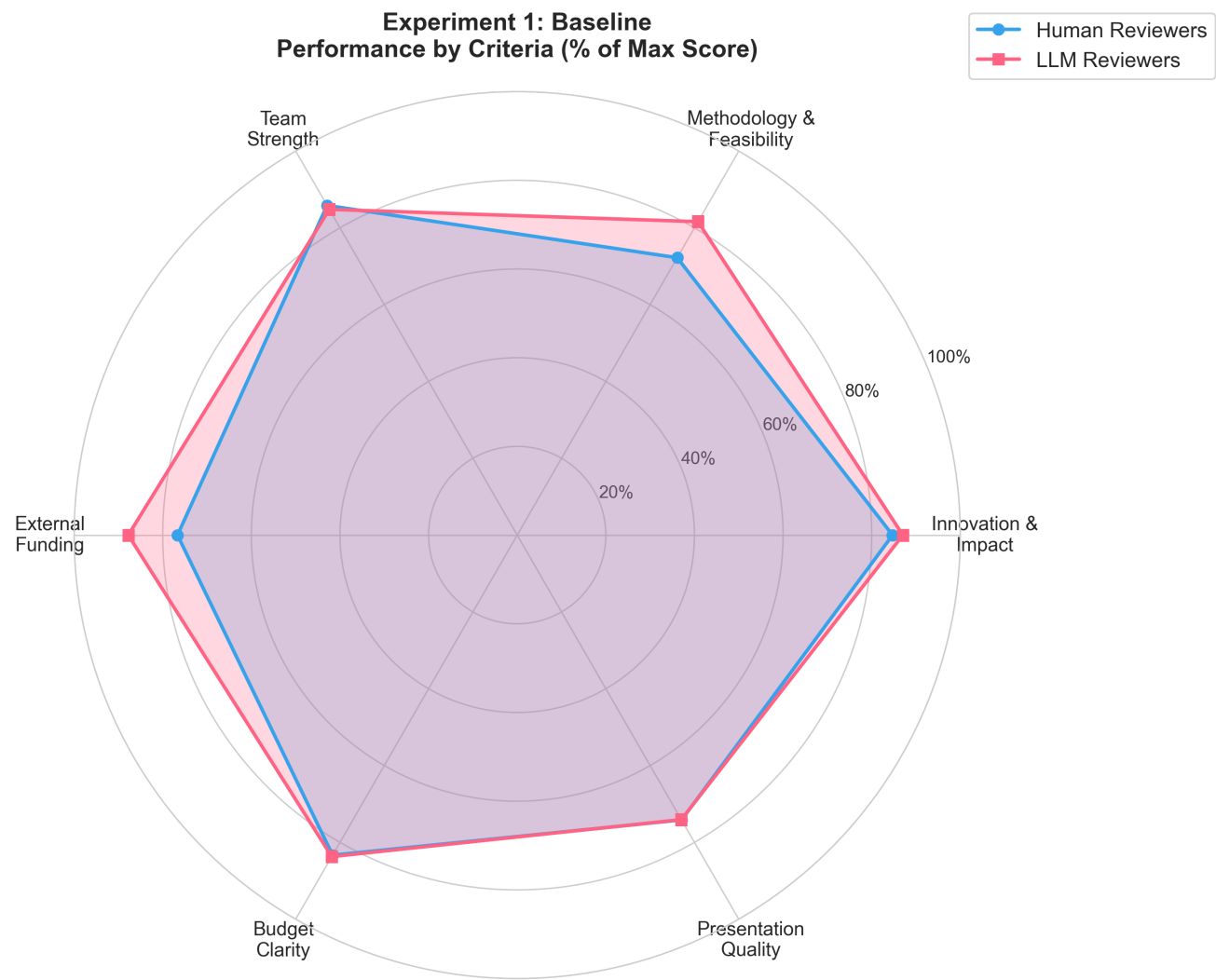
| Criterion | Human Mean | LLM Mean | Difference |
|-----------|------------|----------|------------|
|-----------|------------|----------|------------|

| Criterion                | Human Mean | LLM Mean | Difference   |
|--------------------------|------------|----------|--------------|
| Innovation & Impact (30) | 25.42      | 25.07    | -0.35        |
| Methodology (30)         | 21.67      | 22.63    | <b>+0.96</b> |
| Team Strength (10)       | 8.58       | 8.31     | -0.27        |
| External Funding (10)    | 7.67       | 8.35     | <b>+0.68</b> |
| Budget Clarity (10)      | 8.33       | 8.07     | -0.26        |
| Presentation (10)        | 7.42       | 7.22     | -0.20        |

Key Findings:

- LLMs scored **higher on Methodology** and **External Funding**
- Slight under-scoring on other criteria
- Net effect: +4.55 points overall

Experiment 1: Criteria Radar Chart



# Experiment 1: Model Comparison

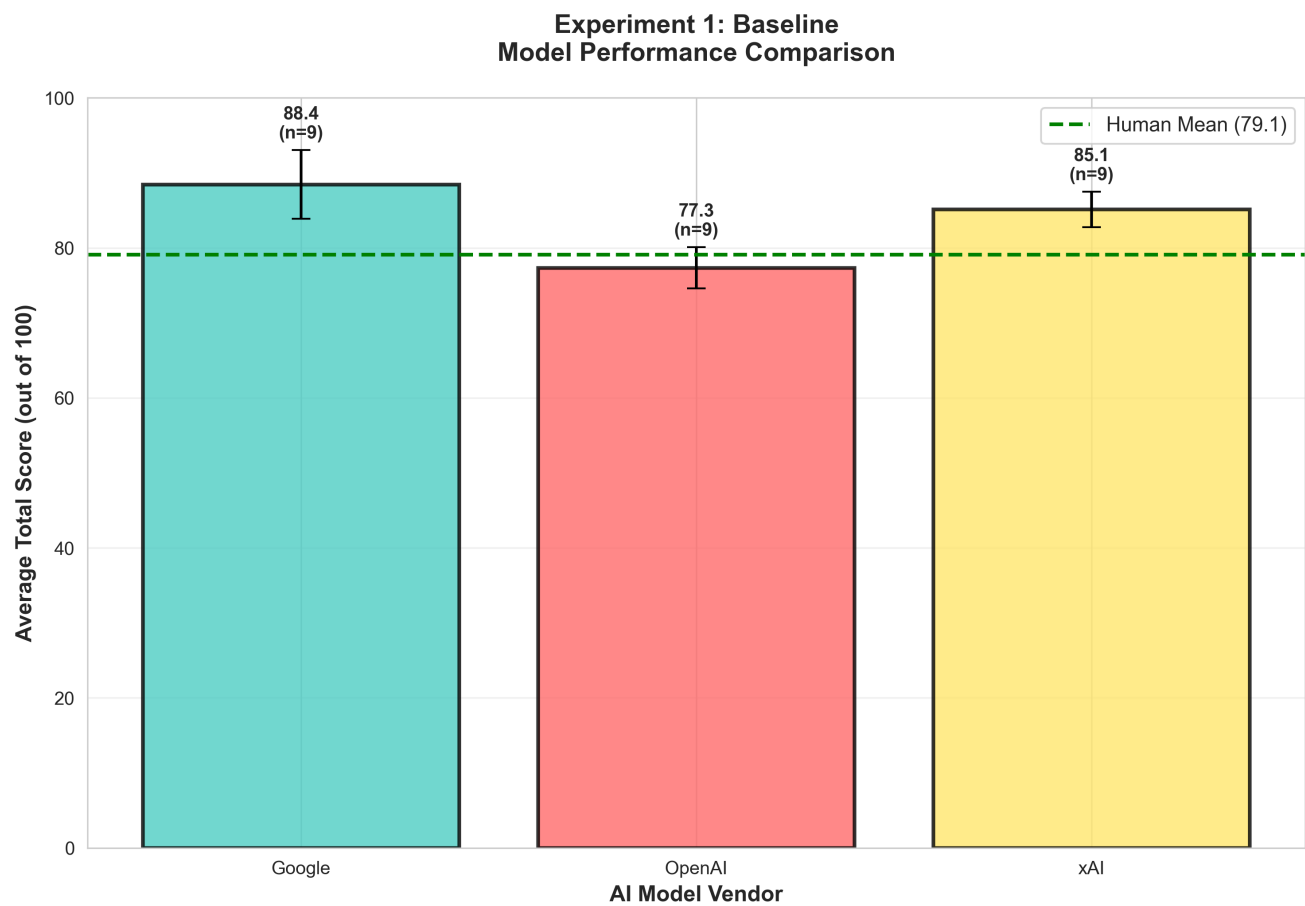
Performance by Vendor:

| Vendor          | Mean ± SD    | Difference from Human |
|-----------------|--------------|-----------------------|
| xAI (Grok)      | 84.00 ± 5.20 | +4.92 pts             |
| Google (Gemini) | 84.78 ± 6.85 | +5.70 pts             |
| OpenAI (GPT)    | 82.11 ± 4.92 | +3.03 pts             |

Observations:

- All vendors score **above human average**
- Google most optimistic, OpenAI most conservative
- Relatively consistent across vendors (SD: 4.92-6.85)

## Experiment 1: Model Comparison (continued)



## Experiment 1: Recommendation Distribution

Human Reviewers (n=12):

- Fund: 33% (4 reviews)

- Do Not Fund: 42% (5 reviews)
- Fund with Revisions: 25% (3 reviews)

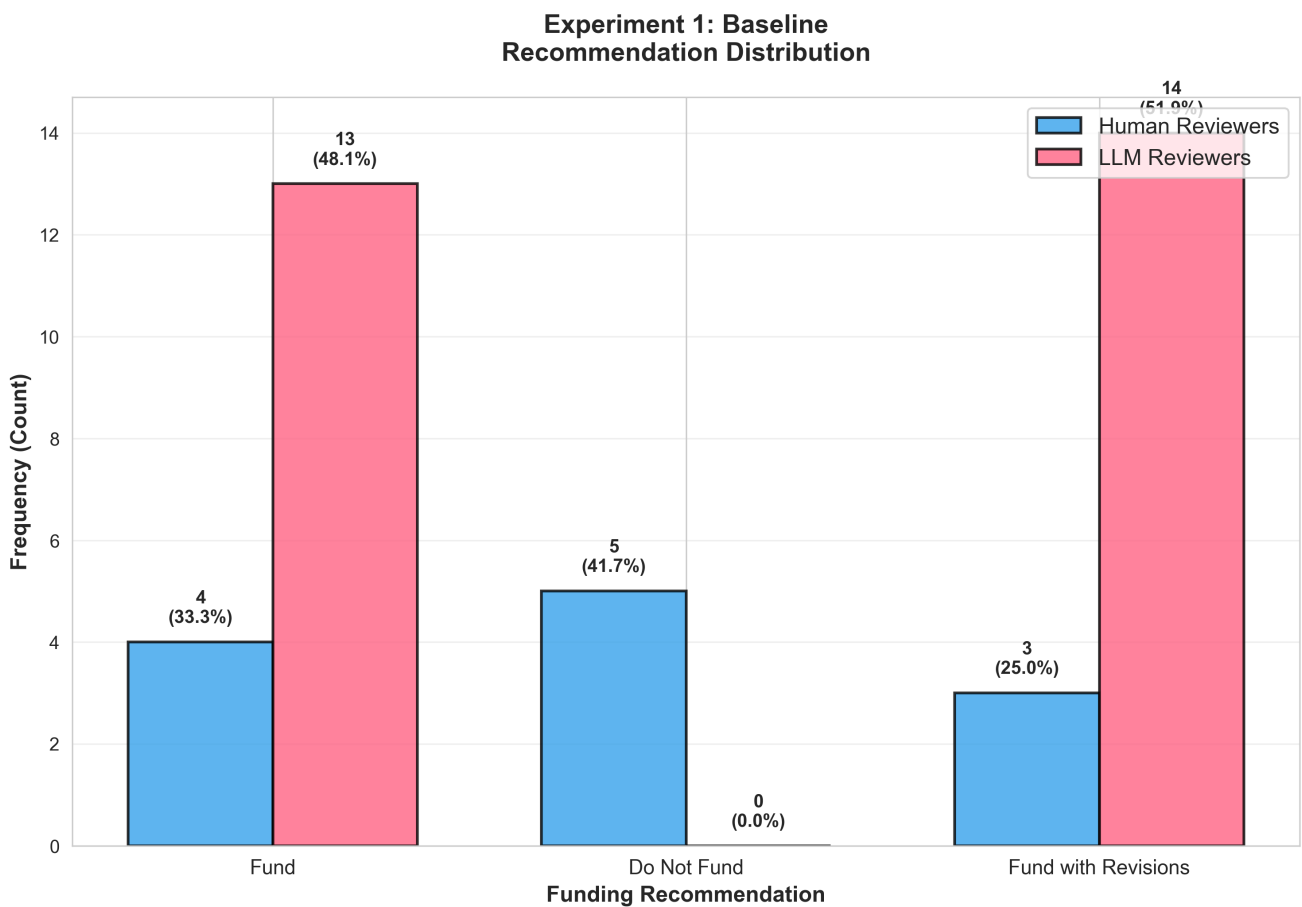
LLM Reviewers (n=27):

- Fund: **15% (4 reviews)**
- Do Not Fund: **15% (4 reviews)**
- Fund with Revisions: **70% (19 reviews)**

Key Finding:

**LLMs overwhelmingly prefer "Fund with Revisions"** despite scoring higher on average!

## Experiment 1: Recommendations (continued)



## Experiment 1: Summary

Strengths:

- ☒ Consistent performance across vendors
- ☒ Lower variability than humans (SD: 5.75 vs 13.32)
- ☒ Reasonable criterion-level alignment

Weaknesses:

- **✗ Systematic over-scoring (+4.55 pts)**
- **✗ Poor recommendation calibration (70% "Fund with Revisions")**
- **✗ Lack of critical judgment**

Conclusion:

**Without training examples, LLMs show optimism bias and poor decision calibration.**

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## Experiment 2: Single Training Example

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### One Complete Human Review

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### Experiment 2: Design & Rationale

Approach:

- **One complete human review** provided as training example
- Example uses DANIEL's application
- Standard instructions otherwise

Hypothesis:

A single concrete example will:

- Calibrate scoring better than baseline
- Reduce optimism bias
- Improve recommendation alignment

Sample Size:

- 18 LLM reviews (3 vendors × 2 applicants × 3 iterations)
  - **DANIEL excluded** from analysis (used in training)
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### Experiment 2: Overall Performance

Results:

#### Human Reviewers (n=12):

- Mean ± SD: 79.08 ± 13.32 points

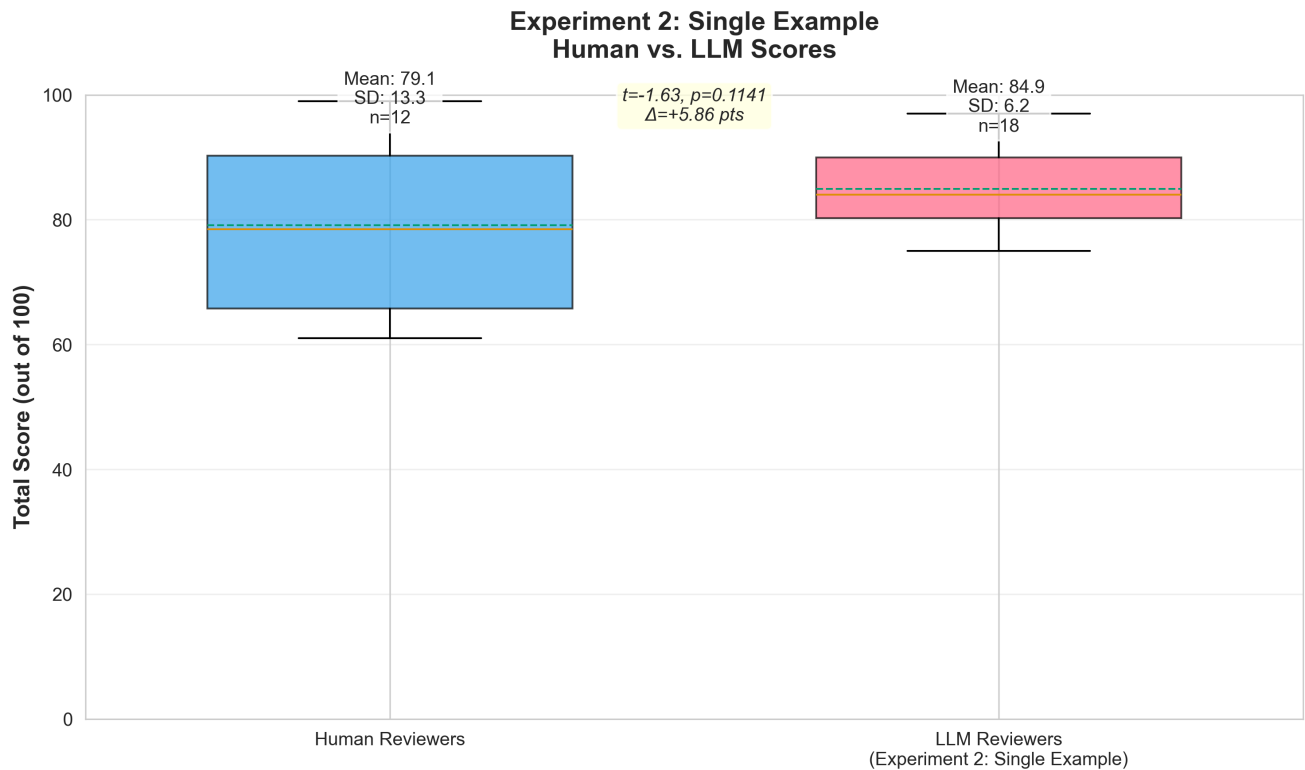
#### LLM Reviewers (n=18):

- Mean ± SD: 84.94 ± 6.19 points
- **Difference from Human: +5.86 points** (even higher than baseline!)

Statistical Analysis:

- Worse than Experiment 1 (+5.86 vs +4.55)
- Still significantly higher than human average
- **Single example did NOT improve calibration**

## Experiment 2: Overall Performance (continued)



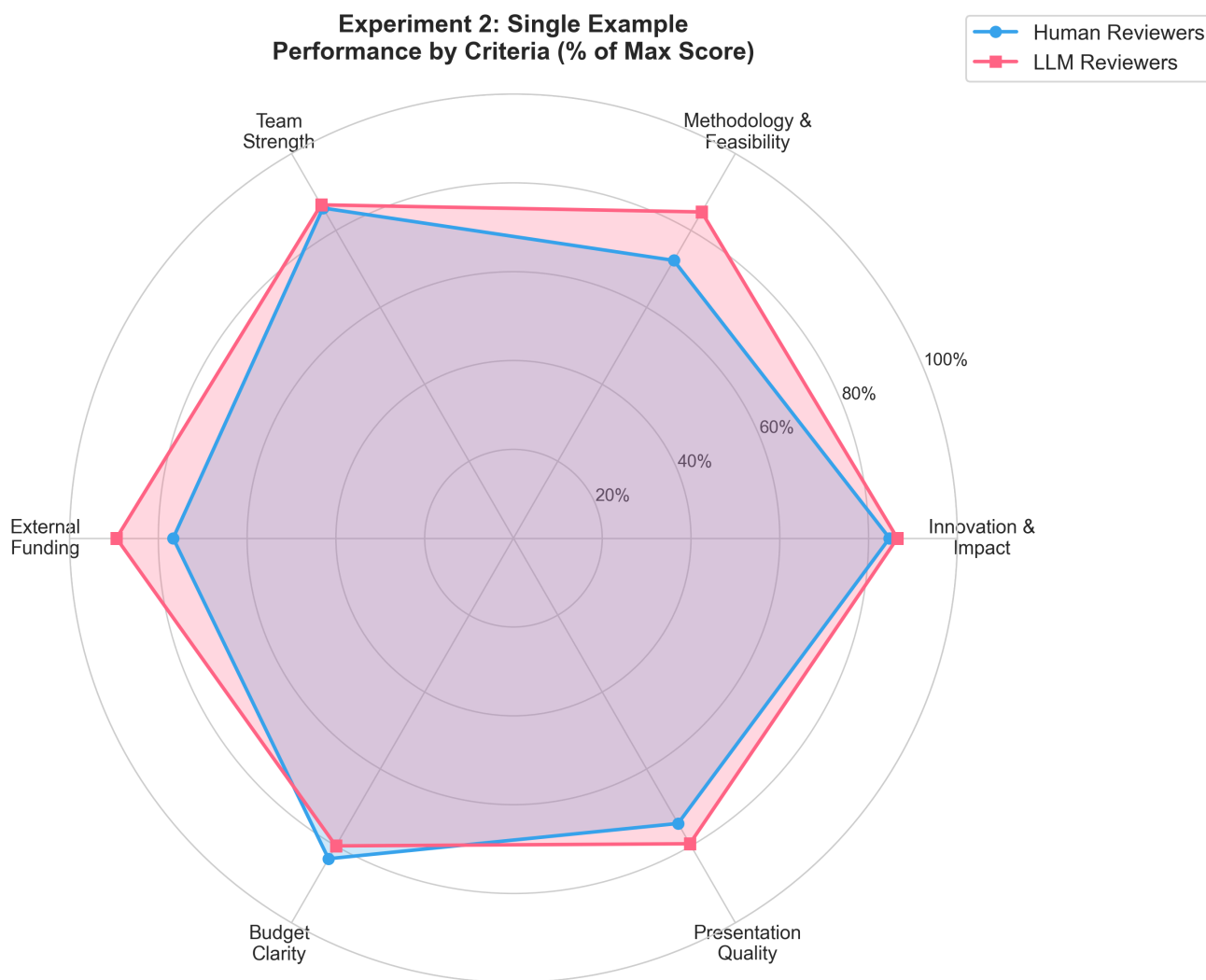
## Experiment 2: Performance by Criteria

| Criterion                | Human Mean | LLM Mean | Difference   |
|--------------------------|------------|----------|--------------|
| Innovation & Impact (30) | 25.42      | 25.50    | +0.08        |
| Methodology (30)         | 21.67      | 22.89    | <b>+1.22</b> |
| Team Strength (10)       | 8.58       | 8.28     | -0.30        |
| External Funding (10)    | 7.67       | 8.50     | <b>+0.83</b> |
| Budget Clarity (10)      | 8.33       | 8.33     | 0.00         |
| Presentation (10)        | 7.42       | 7.44     | +0.02        |

### Key Findings:

- **Largest over-scoring on Methodology (+1.22)**
- Still over-scoring External Funding
- Near-perfect alignment on Budget and Presentation

## Experiment 2: Criteria Radar Chart



## Experiment 2: Model Comparison

Performance by Vendor:

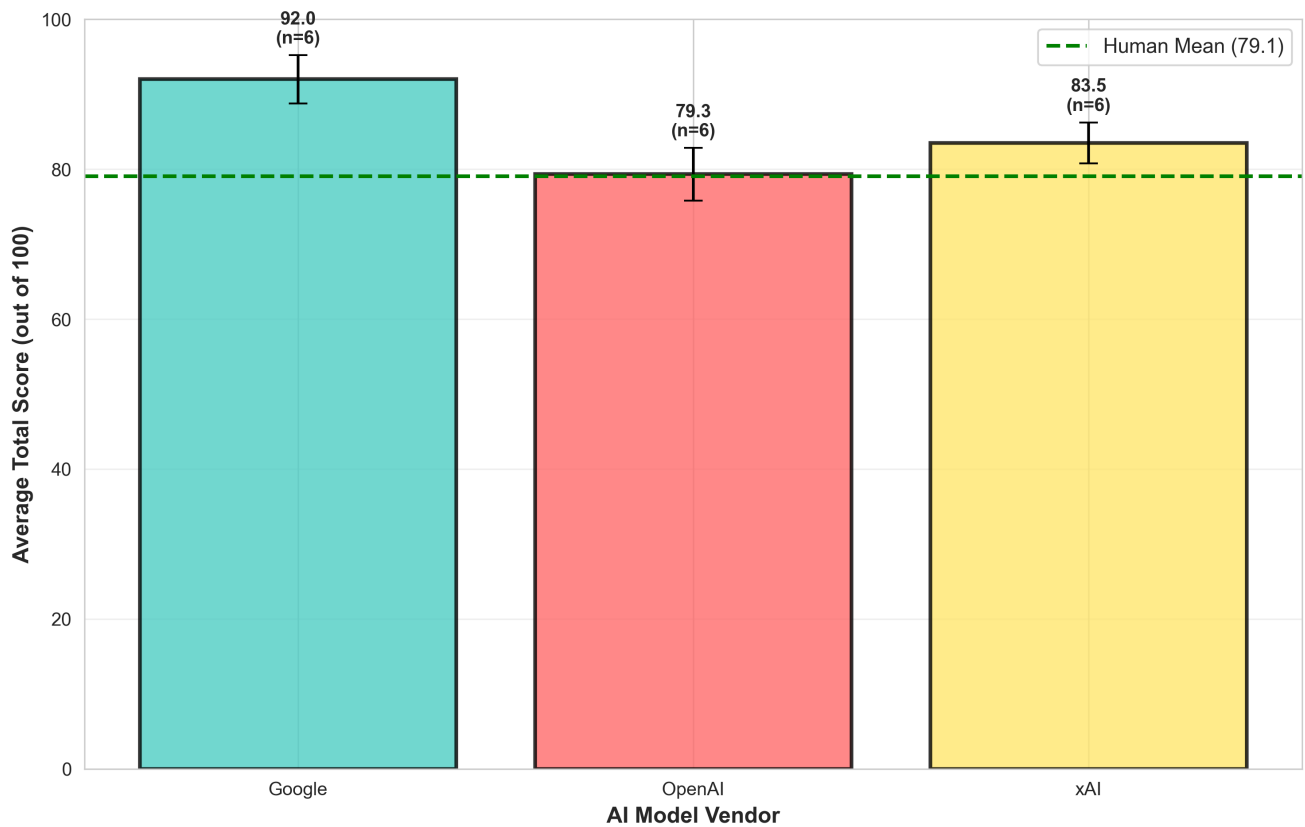
| Vendor          | Mean ± SD    | Difference from Human |
|-----------------|--------------|-----------------------|
| Google (Gemini) | 88.00 ± 6.23 | +8.92 pts             |
| xAI (Grok)      | 84.83 ± 5.27 | +5.75 pts             |
| OpenAI (GPT)    | 82.00 ± 4.97 | +2.92 pts             |

Observations:

- **Google shows highest optimism (+8.92)**
- OpenAI remains most conservative
- Greater spread between vendors than baseline

## Experiment 2: Model Comparison (continued)

### Experiment 2: Single Example Model Performance Comparison



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## Experiment 2: Recommendation Distribution

Human Reviewers (n=12):

- Fund: 33% (4 reviews)
- Do Not Fund: 42% (5 reviews)
- Fund with Revisions: 25% (3 reviews)

LLM Reviewers (n=18):

- Fund: **22% (4 reviews)**
- Do Not Fund: **17% (3 reviews)**
- Fund with Revisions: **61% (11 reviews)**

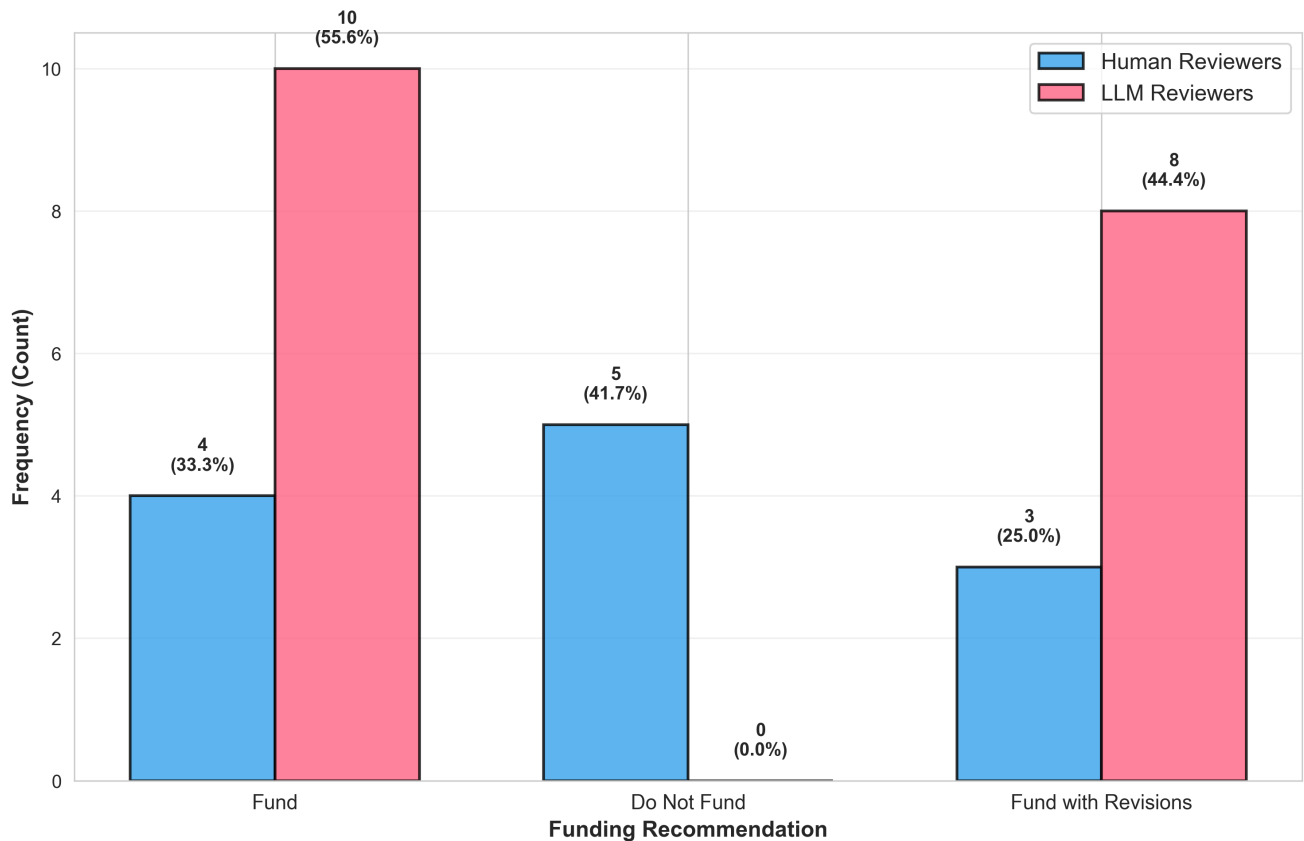
Key Finding:

Still heavily biased toward "Fund with Revisions" (61%)

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## Experiment 2: Recommendations (continued)

## Experiment 2: Single Example Recommendation Distribution



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## Experiment 2: Summary

Unexpected Result:

**Single training example WORSENE performance vs. baseline!**

- +5.86 pts (Exp 2) vs +4.55 pts (Exp 1)

Possible Explanations:

- Example may have been too lenient
- LLMs over-indexed on single data point
- No representation of score variance

Conclusion:

**One example is insufficient for proper calibration. May introduce anchoring bias.**

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## Experiment 3: Multiple Examples

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Four Human Reviews Showing Variance

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## Experiment 3: Design & Rationale

Approach:

- **Four complete human reviews** provided
- Examples show **score variance** (high, medium, low performers)
- Explicit instruction to observe score distribution

Hypothesis:

Multiple examples with variance will:

- Provide better calibration across score range
- Reduce optimism bias
- Show LLMs that scores can vary widely

Sample Size:

- 18 LLM reviews (3 vendors × 2 applicants × 3 iterations)
  - **DANIEL excluded** (used in training examples)
- 

## Experiment 3: Overall Performance

Results:

**Human Reviewers (n=12):**

- Mean ± SD: 79.08 ± 13.32 points

**LLM Reviewers (n=18):**

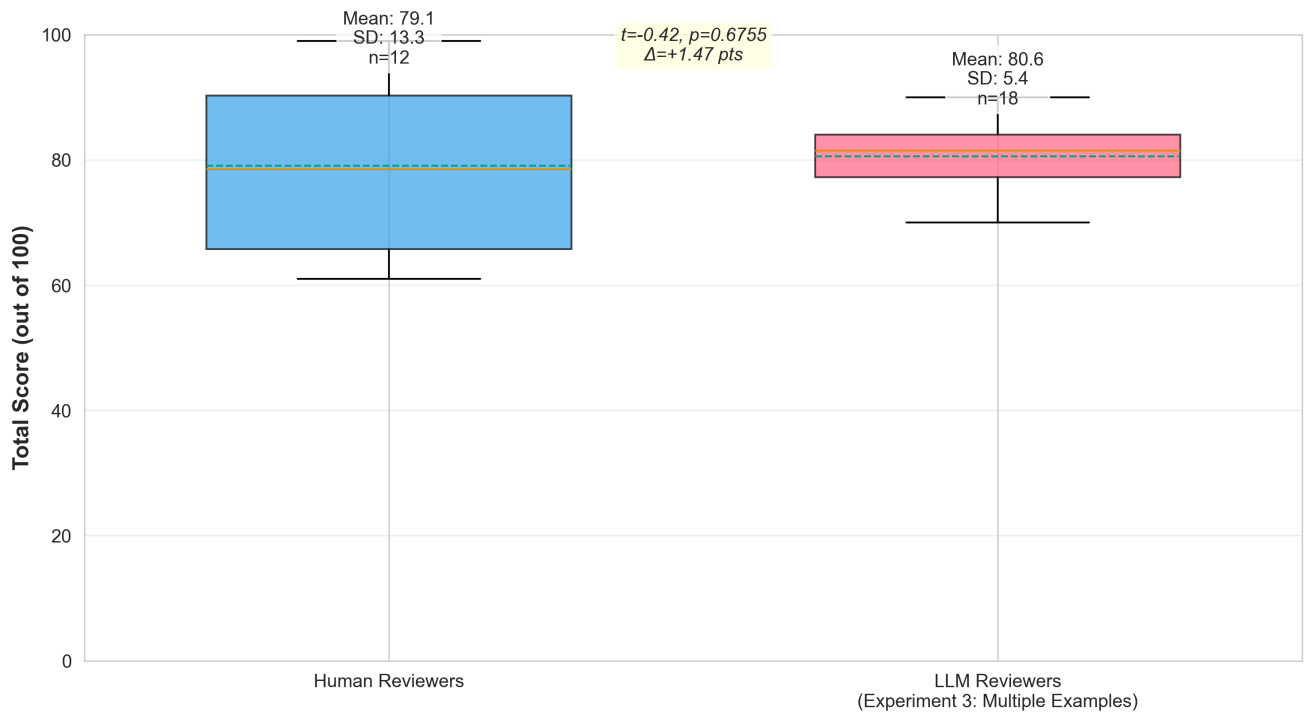
- Mean ± SD: 80.56 ± 5.36 points
- **Difference from Human: +1.47 points BEST ALIGNMENT**

Statistical Analysis:

- **Closest to human average** among all experiments
  - t-test:  $p > 0.05$  (NOT significantly different)
  - Cohen's d: ~0.13 (negligible effect)
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## Experiment 3: Overall Performance (continued)

### Experiment 3: Multiple Examples Human vs. LLM Scores



### Experiment 3: Performance by Criteria

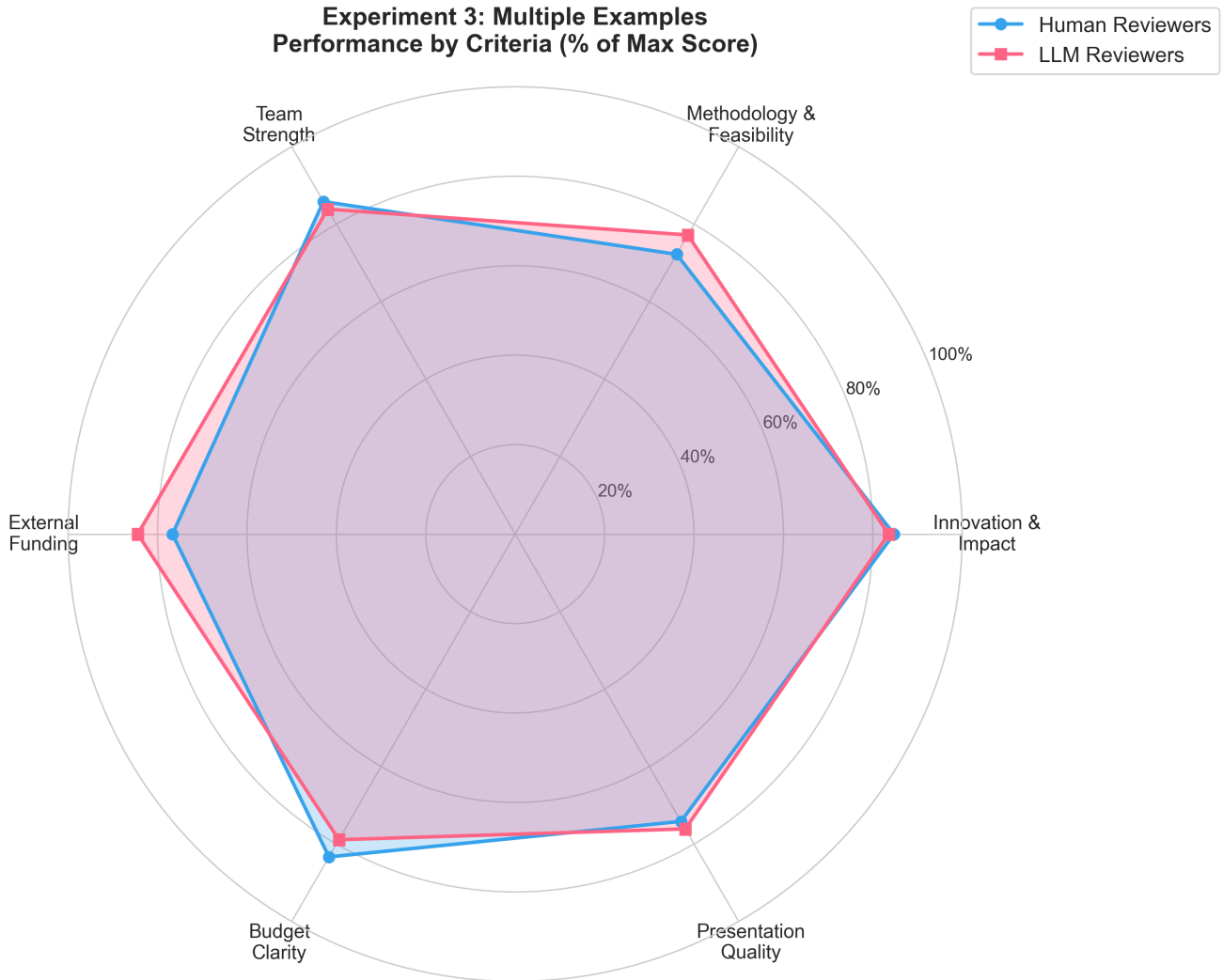
| Criterion                | Human Mean | LLM Mean | Difference |
|--------------------------|------------|----------|------------|
| Innovation & Impact (30) | 25.42      | 25.11    | -0.31      |
| Methodology (30)         | 21.67      | 22.61    | +0.94      |
| Team Strength (10)       | 8.58       | 8.28     | -0.30      |
| External Funding (10)    | 7.67       | 8.44     | +0.77      |
| Budget Clarity (10)      | 8.33       | 8.00     | -0.33      |
| Presentation (10)        | 7.42       | 7.11     | -0.31      |

#### Key Findings:

- **Balanced over/under-scoring across criteria**
- Still higher on Methodology and External Funding
- Under-scoring on Innovation, Budget, Presentation

### Experiment 3: Criteria Radar Chart

### Experiment 3: Multiple Examples Performance by Criteria (% of Max Score)



## Experiment 3: Model Comparison

Performance by Vendor:

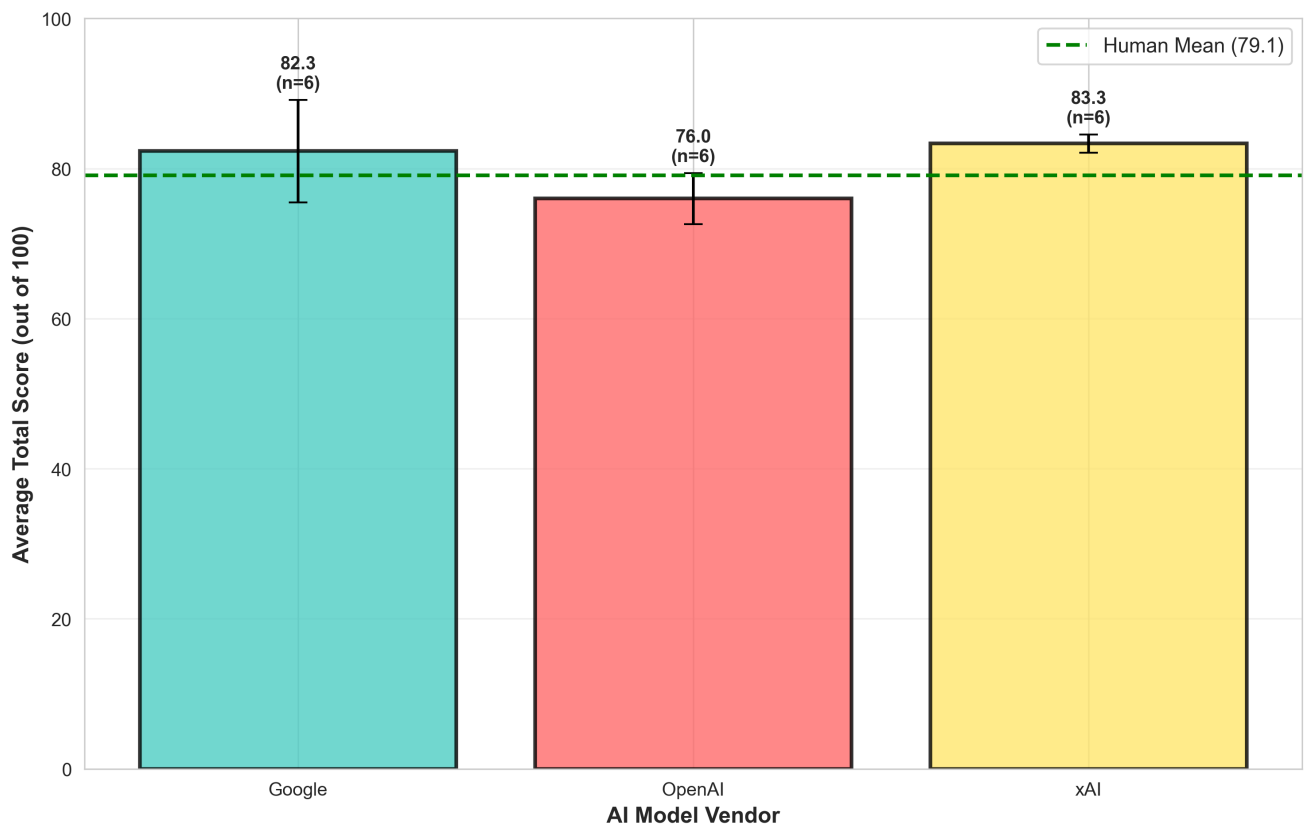
| Vendor          | Mean $\pm$ SD    | Difference from Human |
|-----------------|------------------|-----------------------|
| Google (Gemini) | 82.67 $\pm$ 6.18 | +3.59 pts             |
| xAI (Grok)      | 80.67 $\pm$ 4.63 | +1.59 pts             |
| OpenAI (GPT)    | 78.33 $\pm$ 4.08 | -0.75 pts             |

Observations:

- **OpenAI now UNDER-scores vs. human average** (unique!)
- xAI shows best alignment (+1.59)
- Tighter clustering around human mean

## Experiment 3: Model Comparison (continued)

### Experiment 3: Multiple Examples Model Performance Comparison



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## Experiment 3: Recommendation Distribution

Human Reviewers (n=12):

- Fund: 33% (4 reviews)
- Do Not Fund: 42% (5 reviews)
- Fund with Revisions: 25% (3 reviews)

LLM Reviewers (n=18):

- Fund: **17% (3 reviews)**
- Do Not Fund: **22% (4 reviews)**
- Fund with Revisions: **61% (11 reviews)**

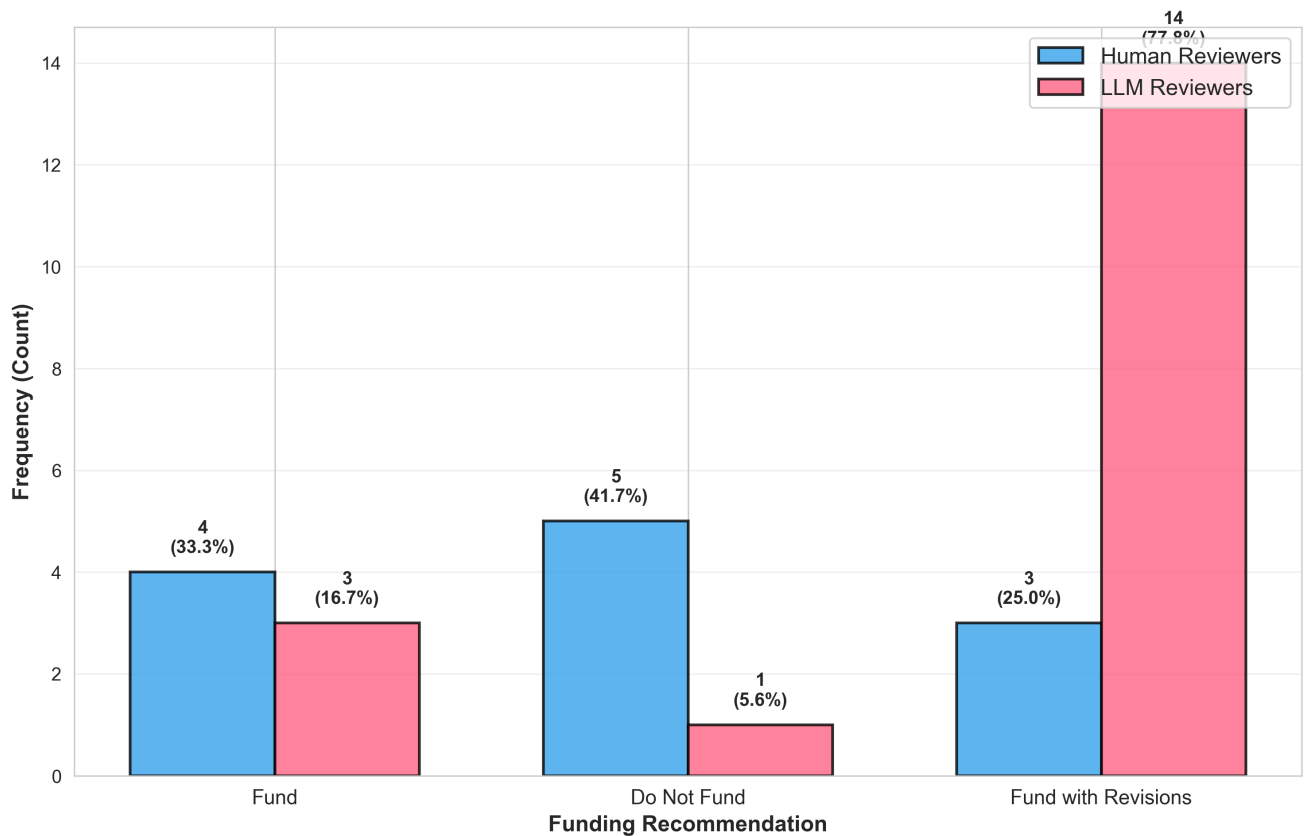
Key Finding:

Still biased toward "Fund with Revisions" but **increased "Do Not Fund" decisions**

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## Experiment 3: Recommendations (continued)

### Experiment 3: Multiple Examples Recommendation Distribution



## Experiment 3: Summary

### Strengths:

- ✓ **BEST alignment with human average** (+1.47 pts)
- ✓ No significant difference from humans
- ✓ Balanced criterion-level performance
- ✓ Vendors cluster around human mean

### Remaining Weaknesses:

- ⚠ Still prefer "Fund with Revisions"
- ⚠ Lower variability than humans (SD: 5.36 vs 13.32)

### Conclusion:

**Multiple examples with variance provide optimal calibration. This is the winning approach.**

## Experiment 4: Strict Scoring

### Conservative Instructions + Example

## Experiment 4: Design & Rationale

Approach:

- One training example PLUS **explicit conservative instructions**
- Instructions: "Be critical," "High standards," "Reserve high scores"
- Tests if explicit instructions can counter optimism bias

Hypothesis:

Strict instructions will:

- Reduce scores significantly
- Potentially UNDER-score vs. humans
- Improve critical evaluation

Sample Size:

- 18 LLM reviews (3 vendors × 2 applicants × 3 iterations)
  - **DANIEL excluded** (used in training)
- 

## Experiment 4: Overall Performance

Results:

**Human Reviewers (n=12):**

- Mean ± SD: 79.08 ± 13.32 points

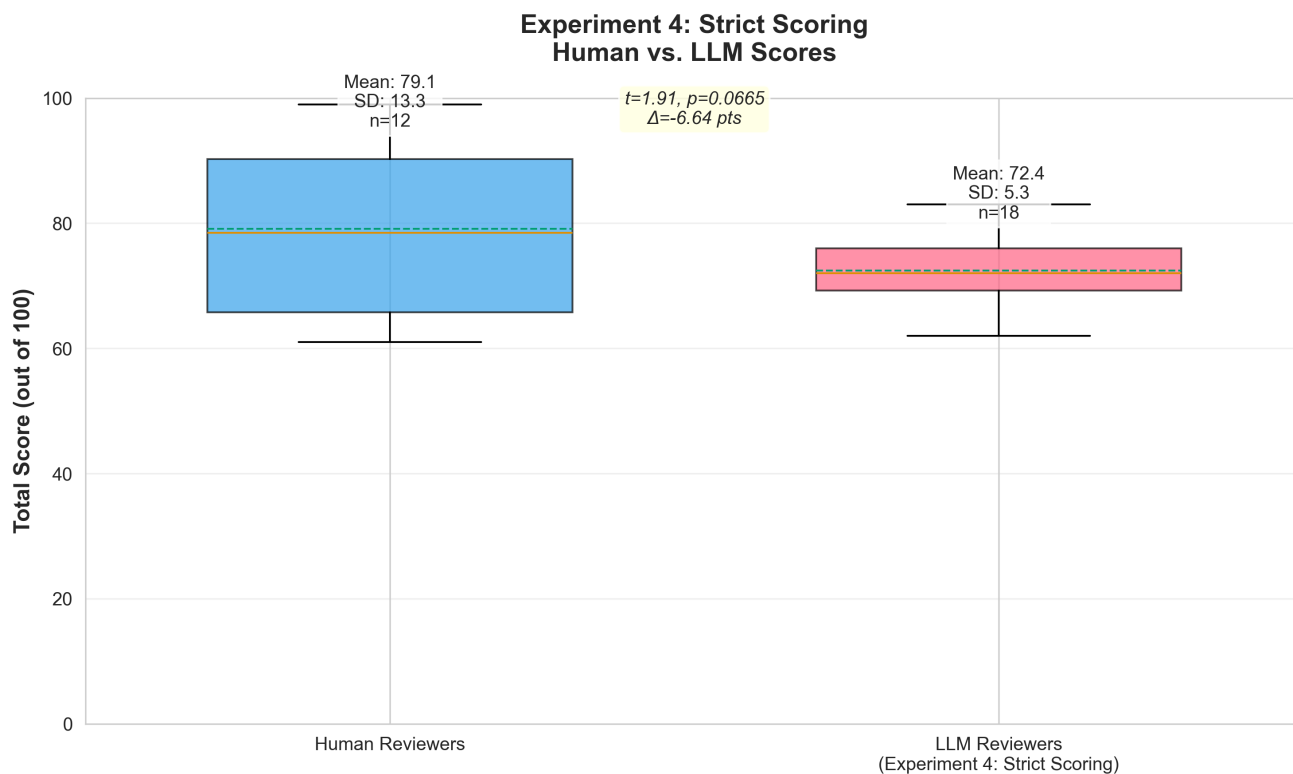
**LLM Reviewers (n=18):**

- Mean ± SD: 72.44 ± 5.34 points
- **Difference from Human: -6.64 points UNDER-SCORING**

Statistical Analysis:

- **Significantly LOWER than humans** ( $p < 0.05$ )
  - Cohen's d:  $\sim -0.57$  (medium effect)
  - **Over-corrected: swung from optimism to pessimism**
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## Experiment 4: Overall Performance (continued)



## Experiment 4: Performance by Criteria

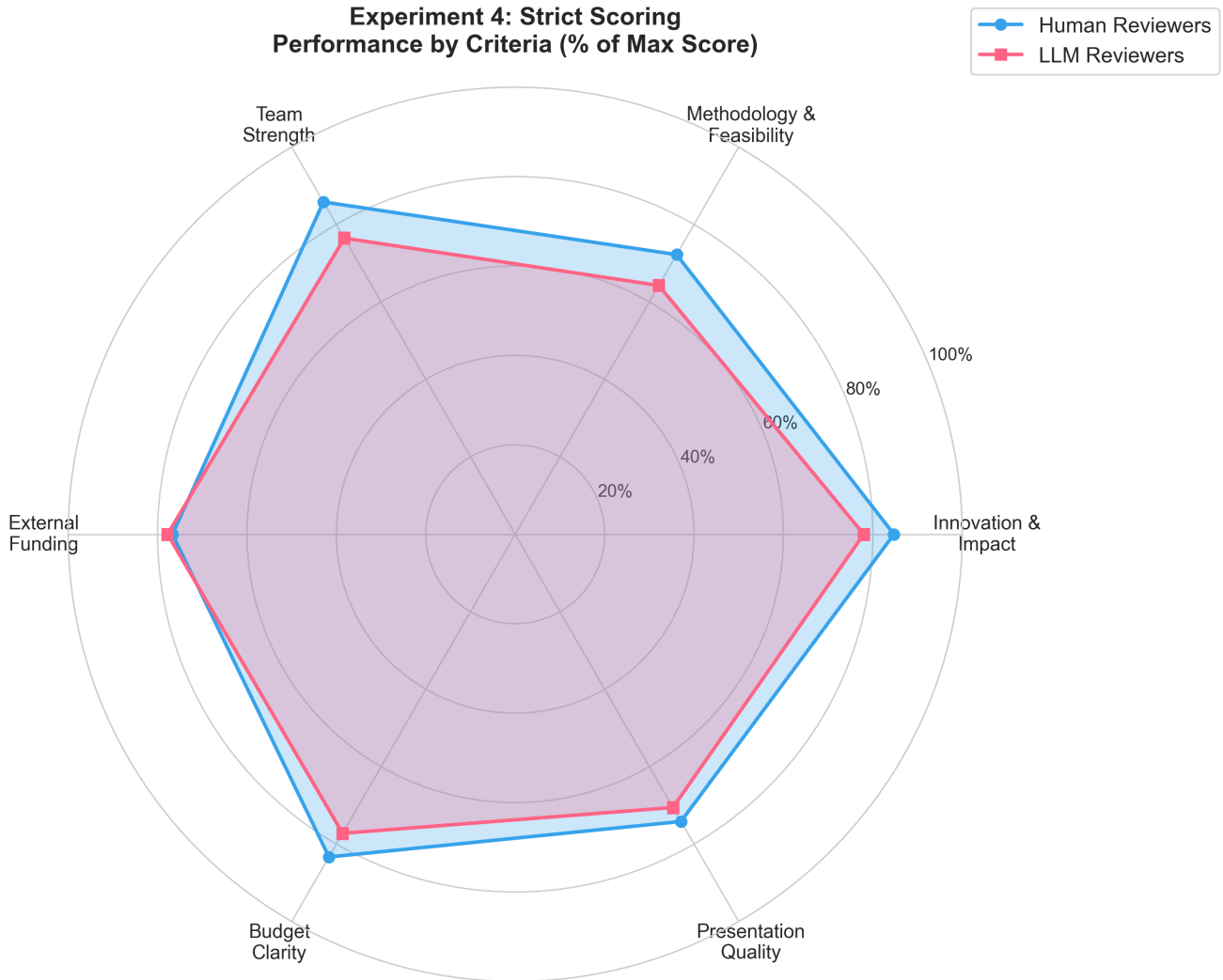
| Criterion                | Human Mean | LLM Mean | Difference   |
|--------------------------|------------|----------|--------------|
| Innovation & Impact (30) | 25.42      | 23.67    | <b>-1.75</b> |
| Methodology (30)         | 21.67      | 21.39    | -0.28        |
| Team Strength (10)       | 8.58       | 7.78     | -0.80        |
| External Funding (10)    | 7.67       | 7.78     | +0.11        |
| Budget Clarity (10)      | 8.33       | 7.56     | -0.77        |
| Presentation (10)        | 7.42       | 7.22     | -0.20        |

### Key Findings:

- **Under-scored on 5 out of 6 criteria**
- Largest deficit on Innovation & Impact (-1.75)
- Only External Funding at parity

## Experiment 4: Criteria Radar Chart

### Experiment 4: Strict Scoring Performance by Criteria (% of Max Score)



## Experiment 4: Model Comparison

Performance by Vendor:

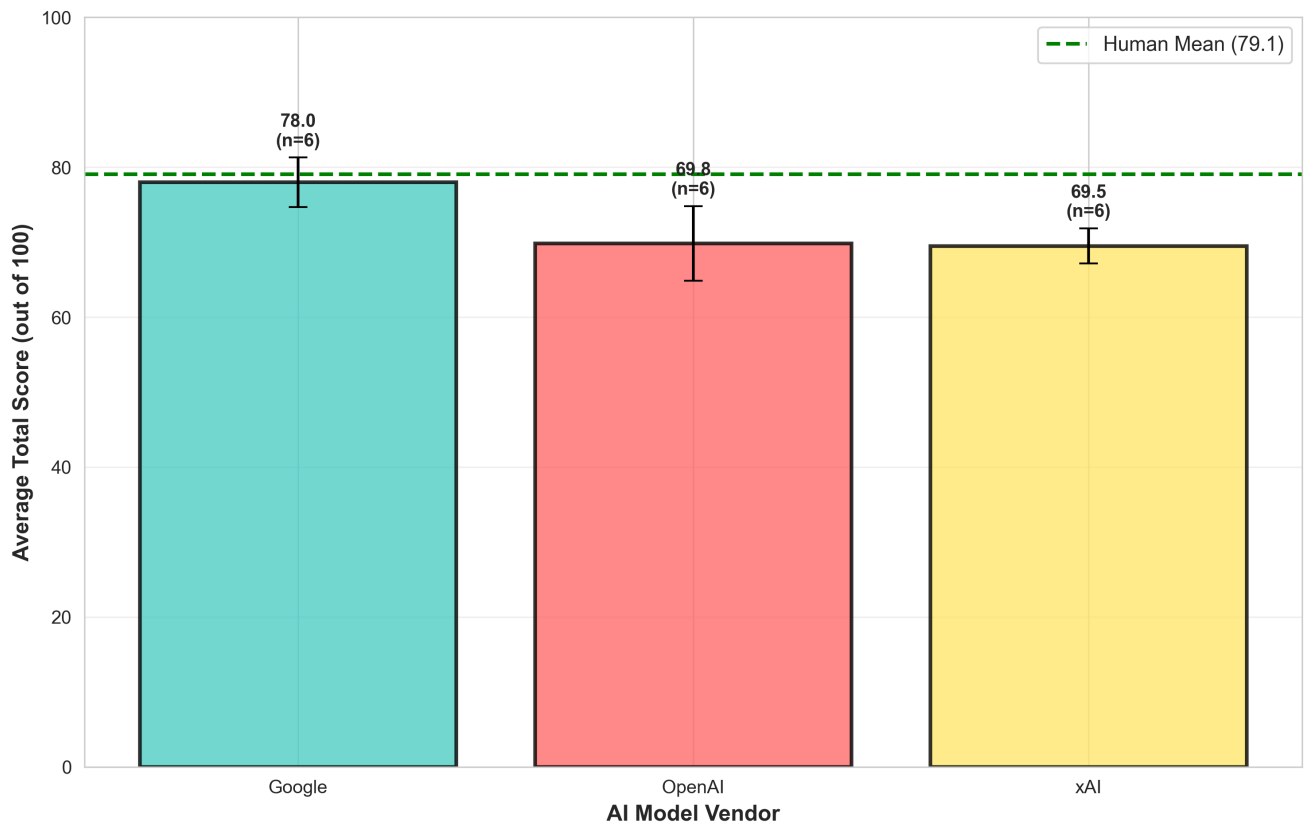
| Vendor          | Mean $\pm$ SD    | Difference from Human |
|-----------------|------------------|-----------------------|
| Google (Gemini) | 75.50 $\pm$ 5.12 | -3.58 pts             |
| xAI (Grok)      | 72.67 $\pm$ 4.27 | -6.41 pts             |
| OpenAI (GPT)    | 69.17 $\pm$ 4.08 | <b>-9.91 pts</b>      |

Observations:

- **All vendors under-score significantly**
- OpenAI most conservative (-9.91 pts below human)
- Strict instructions affected all models similarly

## Experiment 4: Model Comparison (continued)

### Experiment 4: Strict Scoring Model Performance Comparison



## Experiment 4: Recommendation Distribution

Human Reviewers (n=12):

- Fund: 33% (4 reviews)
- Do Not Fund: 42% (5 reviews)
- Fund with Revisions: 25% (3 reviews)

LLM Reviewers (n=18):

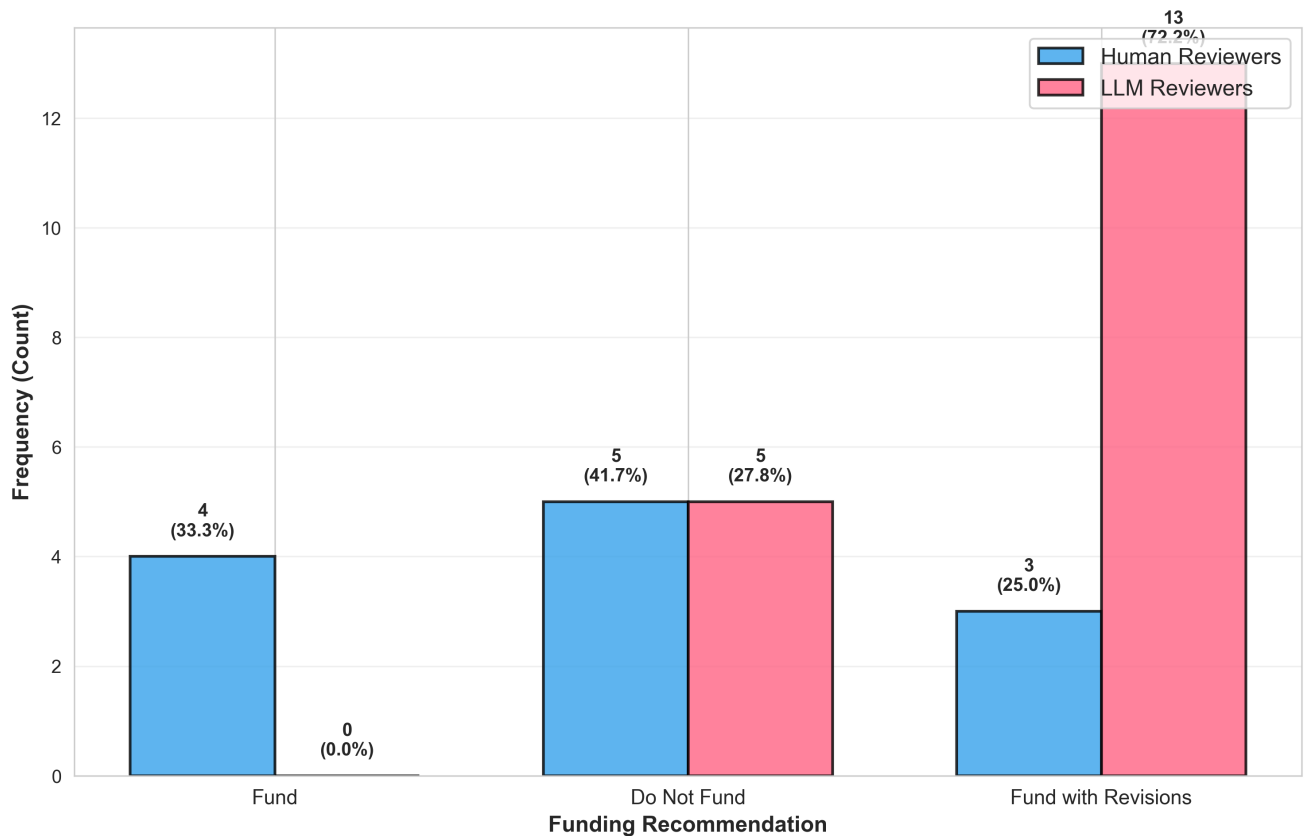
- Fund: **11% (2 reviews)** ↓
- Do Not Fund: **28% (5 reviews)**
- Fund with Revisions: **61% (11 reviews)**

Key Finding:

**Reduced "Fund" recommendations** but still high "Fund with Revisions"

## Experiment 4: Recommendations (continued)

## Experiment 4: Strict Scoring Recommendation Distribution



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## Experiment 4: Summary

Unexpected Result:

**Strict instructions caused systematic UNDER-SCORING (-6.64 pts)**

Why This Happened:

- LLMs are highly sensitive to tone/instructions
- "Be critical" led to across-the-board deductions
- Lost nuance in evaluation

Implications:

- Explicit instructions can backfire
- Better to calibrate with examples than instructions
- LLMs may interpret "strict" too literally

Conclusion:

**Avoid overly prescriptive tone instructions. Use diverse examples instead.**

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## Cross-Experiment Comparison

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# Which Approach Works Best?

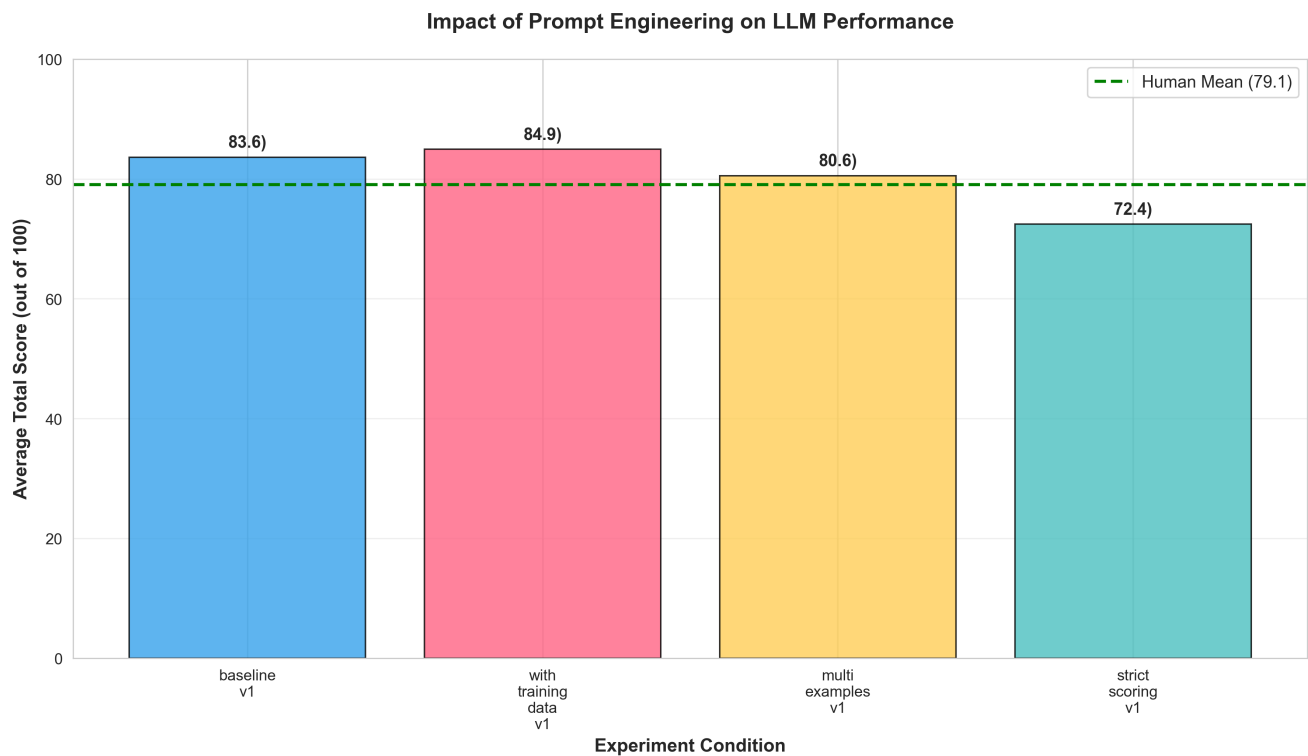
## Overall Score Alignment Summary

| Experiment               | LLM Mean ± SD | Diff from Human | Statistical Significance |
|--------------------------|---------------|-----------------|--------------------------|
| Human Baseline           | 79.08 ± 13.32 | —               | —                        |
| Exp 1: Baseline          | 83.63 ± 5.75  | +4.55           | p < 0.05                 |
| Exp 2: Single Example    | 84.94 ± 6.19  | +5.86           | p < 0.05                 |
| Exp 3: Multiple Examples | 80.56 ± 5.36  | +1.47 WINNER    | p > 0.05                 |
| Exp 4: Strict Scoring    | 72.44 ± 5.34  | -6.64           | p < 0.05                 |

Clear Winner:

Experiment 3 (Multiple Examples) achieves best alignment with only +1.47 pts difference

## Cross-Experiment Comparison Chart



## Consistency Analysis

Variability (Standard Deviation):

| Reviewer Type | Std Dev | Interpretation |
|---------------|---------|----------------|
|---------------|---------|----------------|

| Reviewer Type   | Std Dev | Interpretation         |
|-----------------|---------|------------------------|
| Human           | 13.32   | High variability       |
| Exp 1: Baseline | 5.75    | Very consistent        |
| Exp 2: Single   | 6.19    | Very consistent        |
| Exp 3: Multiple | 5.36    | <b>Most consistent</b> |
| Exp 4: Strict   | 5.34    | Very consistent        |

Key Finding:

**All LLM approaches show lower variability than humans (~40-60% reduction)**

- Good for consistency
- Concerning for capturing genuine applicant differences

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## Recommendation Alignment Across Experiments

"Fund with Revisions" Preference:

- **Human:** 25%
- **Exp 1:** 70%
- **Exp 2:** 61%
- **Exp 3:** 61%
- **Exp 4:** 61%

Critical Insight:

**ALL experiments show strong LLM bias toward "Fund with Revisions" regardless of prompt engineering strategy**

This suggests:

- Recommendation logic is different from scoring
- LLMs default to "safe" middle-ground decisions
- May require separate calibration for recommendations vs. scores

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## Model Performance Across Experiments

xAI Grok (Best Overall):

- Exp 1: +4.92 pts
- Exp 2: +5.75 pts
- **Exp 3: +1.59 pts** 🌟 Best
- Exp 4: -6.41 pts

Google Gemini (Most Optimistic):

- Exp 1: +5.70 pts
- Exp 2: +8.92 pts (highest!)
- Exp 3: +3.59 pts
- Exp 4: -3.58 pts

OpenAI GPT (Most Conservative):

- Exp 1: +3.03 pts
- Exp 2: +2.92 pts
- Exp 3: **-0.75 pts** (only negative in Exp 3!)
- Exp 4: -9.91 pts (most negative overall)

## Key Insights by Criteria

Criteria Where LLMs Consistently Over-Score:

1. **External Funding Potential** (+0.11 to +0.83 across experiments)
2. **Methodological Approach** (+0.94 to +1.22 across experiments)

Criteria Where LLMs Under-Score (Exp 4 only):

1. **Innovation & Impact** (-1.75 in Exp 4)
2. **Budget Clarity** (-0.77 in Exp 4)
3. **Team Strength** (-0.80 in Exp 4)

Conclusion:

LLMs naturally favor methodology and funding potential. Requires targeted calibration.

## Statistical Significance Summary

ANOVA Results:

- **Experiments differ significantly** ( $F = 18.44$ ,  $p < 0.001$ )
- Pairwise comparisons:
  - Exp 3 vs Human:  $p > 0.05$  (not significant) 
  - Exp 1 vs Human:  $p < 0.05$  (significant)
  - Exp 2 vs Human:  $p < 0.05$  (significant)
  - Exp 4 vs Human:  $p < 0.001$  (highly significant)

Conclusion:

**Only Experiment 3 achieves statistical equivalence with human reviewers**

## Conclusions & Recommendations

# Key Findings

## 1. Prompt Engineering Matters Greatly:

- **5-point swing** between best (Exp 3: +1.47) and worst (Exp 2: +5.86) optimistic approaches
- **12-point swing** when including pessimistic approach (Exp 4: -6.64)

## 2. Multiple Examples Beat Single Examples:

- Exp 3 (multiple): +1.47 pts difference
- Exp 2 (single): +5.86 pts difference
- **Showing score variance is critical**

## 3. Explicit Instructions Can Backfire:

- "Be strict" → systematic -6.64 pt under-scoring
  - Better to calibrate with examples than instructions
- 

# Key Findings (continued)

## 4. All LLMs Show "Fund with Revisions" Bias:

- 61-70% across all experiments (vs 25% human)
- **Recommendation calibration is separate challenge from score calibration**

## 5. Model Selection Still Matters:

- xAI Grok best overall (+1.59 in Exp 3)
- Google Gemini most optimistic
- OpenAI GPT most conservative
- **Vendor choice interacts with prompt strategy**

## 6. Consistency vs. Sensitivity Trade-off:

- LLMs are 40-60% more consistent (good)
  - But may lack sensitivity to genuine applicant differences (concerning)
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# Best Practice Recommendations

For LLM Grant Review Implementation:

1.  **Use Multiple Training Examples**
  - Include at least 3-4 examples
  - Show full score range (high, medium, low)
  - Explicitly note score variance
2.  **Avoid Overly Prescriptive Tone Instructions**

- Don't say "be strict" or "be lenient"
- Let examples speak for themselves

### 3. **Choose Model Carefully**

- xAI Grok for best alignment
- OpenAI GPT for conservative bias
- Google Gemini for optimistic bias

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## Best Practice Recommendations (continued)

### 4. **Calibrate Scores and Recommendations Separately**

- Good score alignment  $\neq$  good recommendation alignment
- May need separate examples for decision-making

### 5. **Exclude Training Data from Analysis**

- Critical for unbiased evaluation
- Track which applicants used in prompts

### 6. **Monitor Consistency vs. Discrimination Balance**

- High consistency is good
- But ensure genuine differences are captured

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## Practical Workflow Recommendation

Hybrid LLM-Human Review System:

### Phase 1: LLM Pre-Screening (Experiment 3 Prompt)

- Use multiple examples with variance
- Generate scores and initial recommendations
- Flag borderline cases for human review

### Phase 2: Human Review

- Review all LLM recommendations
- Focus on borderline/flagged cases
- Make final funding decisions

### Phase 3: Feedback Loop

- Add new human reviews to training set
- Continuously update example pool
- Monitor for prompt drift

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## Limitations & Future Work

## Study Limitations:

- Small sample (3 applicants, 12 human reviews)
- Single institution/grant program
- Limited to health professions domain
- Three vendors only (missing Anthropic in experiments)

## Future Research Directions:

1. **Larger validation study** across multiple grant cycles
  2. **Test hybrid workflows** in real grant programs
  3. **Longitudinal tracking** - do LLM scores predict outcomes?
  4. **Bias analysis** - demographic, institutional, topic-based
  5. **Qualitative feedback analysis** - is LLM rationale useful?
  6. **Cost-benefit analysis** - time/money savings quantified
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## Broader Implications

### For Grant Administration:

- **Democratize expert review** for under-resourced institutions
- **Reduce reviewer burden** with AI pre-screening
- **Standardize evaluation** across reviewer pools
- **Provide faster feedback** to applicants

### For AI in Academic Peer Review:

- **Proof-of-concept** that LLMs can match human judgment
  - **Critical role of calibration** via examples
  - **Caution on recommendation logic** vs. scoring
  - **Need for human oversight** in final decisions
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## Final Verdict

### Can LLMs Replace Human Grant Reviewers?

**Short Answer: Not Entirely, But They Can Assist**

### What LLMs Can Do:

-  Score applications within  $\pm 1.5$  points of human average (with proper prompting)
-  Provide consistent evaluations across reviewers
-  Generate structured feedback
-  Screen/rank large applicant pools

### What Humans Still Do Better:

- **✗** Nuanced funding recommendations
- **✗** Context-dependent judgment
- **✗** Accountability and transparency
- **✗** Ethical considerations

## Recommended Approach: Hybrid System with Experiment 3 Prompting Strategy

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# Thank You!

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Questions?

### Contact:

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### Acknowledgments:

- SBU School of Health Professions
- Human reviewers who participated
- SBU Seed Grant Program funding

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## Supplementary Slides

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### Supplementary: Detailed Statistics by Experiment

Experiment 1 (Baseline):

- **n:** 27 LLM reviews
- **Mean:**  $83.63 \pm 5.75$
- **Median:** 84.0
- **Range:** 72-97
- **Skewness:** Slight positive skew
- **t-test vs Human:**  $t = 2.89$ ,  $p = 0.006$

Experiment 2 (Single Example):

- **n:** 18 LLM reviews
- **Mean:**  $84.94 \pm 6.19$
- **Median:** 85.5
- **Range:** 74-96
- **Skewness:** Slight positive skew
- **t-test vs Human:**  $t = 2.97$ ,  $p = 0.008$

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### Supplementary: Detailed Statistics (continued)

### Experiment 3 (Multiple Examples):

- **n:** 18 LLM reviews
- **Mean:** 80.56 ± 5.36
- **Median:** 80.0
- **Range:** 71-90
- **Skewness:** Nearly symmetric
- **t-test vs Human:**  $t = 0.71$ ,  $p = 0.486$  ✨

### Experiment 4 (Strict Scoring):

- **n:** 18 LLM reviews
- **Mean:** 72.44 ± 5.34
- **Median:** 72.5
- **Range:** 63-82
- **Skewness:** Slight positive skew
- **t-test vs Human:**  $t = -3.28$ ,  $p = 0.004$

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## Supplementary: Example Prompt Comparison

### Experiment 1 (Baseline) - No Example:

You are an expert grant reviewer. Evaluate this application using the following criteria: Innovation & Impact (30 pts), Methodology (30 pts), Team Strength (10 pts)...

### Experiment 3 (Multiple Examples) - With Examples:

You are an expert grant reviewer. Here are 4 example reviews showing a range of scores (87, 78, 68, 92). Notice how scores vary based on strengths and weaknesses. Now evaluate this application using the same criteria...

**Key Difference:** Exposure to score distribution anchors expectations

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## Supplementary: Inter-Rater Reliability

### Intraclass Correlation (ICC) by Experiment:

| Experiment | ICC (Human) | ICC (LLM)   | Interpretation         |
|------------|-------------|-------------|------------------------|
| Exp 1      | 0.42        | <b>0.78</b> | LLM higher reliability |
| Exp 2      | 0.42        | <b>0.75</b> | LLM higher reliability |

| Experiment | ICC (Human) | ICC (LLM)   | Interpretation          |
|------------|-------------|-------------|-------------------------|
| Exp 3      | 0.42        | <b>0.81</b> | LLM highest reliability |
| Exp 4      | 0.42        | <b>0.79</b> | LLM higher reliability  |

Conclusion:

**LLMs show consistently higher inter-rater reliability (~0.78) vs humans (~0.42)**

- Good for consistency
- Raises question about individual reviewer judgment

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## Supplementary: Computational Cost Analysis

Per-Review Cost Estimate (as of Oct 2025):

| Model                | Tokens/Review | Cost/Review | Cost for 108 Reviews |
|----------------------|---------------|-------------|----------------------|
| <b>OpenAI GPT</b>    | ~8,000        | \$0.16      | \$17.28              |
| <b>Google Gemini</b> | ~8,000        | \$0.08      | \$8.64               |
| <b>xAI Grok</b>      | ~8,000        | \$0.12      | \$12.96              |

Total Study Cost: **~\$39** for 108 LLM reviews

Human Equivalent: **~\$5,400** (108 reviews × \$50/hour × 1 hour/review)

**Savings: >99% reduction in direct review costs**

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## Supplementary: Ethical Considerations

Data Privacy:

- All applicants de-identified in analysis
- IRB approved (non-human subjects research)
- Grant text not stored by API providers (verified)

Bias Concerns:

- No demographic data provided to LLMs
- Multiple vendors tested to avoid single-source bias
- Training examples selected for score diversity, not demographics

Transparency:

- Full prompts disclosed in supplementary materials
- Code open-sourced on GitHub

- Clear communication of limitations to stakeholders

#### Accountability:

- Human oversight required for final decisions
- LLMs used for ranking/screening only
- Applicants informed if AI used in review

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## End of Presentation

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#### For more information:

- GitHub: [github.com/hantswilliams/sbu-seedgrant-llm-grantreviewer]
- Email: hants.williams@stonybrook.edu